

Therefore the conditional mean
of X given $Y=y$ is

$$E(X|Y=y) = \int_0^y x \cdot f(x|Y) dx$$

$$= \int_0^y x \frac{1}{y} dx = \frac{1}{y} \left[\frac{x^2}{2} \right]_0^y$$

$$= \frac{1}{2y} [y^2 - 0] = \frac{1}{2y} \cdot y^2 = \frac{y}{2}$$

Similarly

$$E(X^2|Y=y) = \int_0^y x^2 f(x|Y) dx$$

$$= \int_0^y x^2 \frac{1}{y} dx = \frac{1}{y} \left[\frac{x^3}{3} \right]_0^y$$

$$= \frac{1}{3} [y^3 - 0] = \frac{y^3}{3}$$

Therefore,

$$\text{Var}(X|Y) = E(X^2|Y) - [E(X|Y)]^2$$

$$= \frac{y^3}{3} - \left[\frac{y}{2} \right]^2$$

$$= \frac{y^3}{3} - \frac{y^2}{4} = \frac{y^2}{12}, \quad 0 < y < 1$$

satisfied.

Example

$$f(x,y) = \begin{cases} 2 & 0 < x < y, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Show that variance of conditional distribution of Y , given $x = X$ is

$$(1-x)^2/12, \quad 0 < x < 1$$

and also of X , given $y = Y$ is $y^2/12, \quad 0 < y < 1$!

Solution:

complicated domain:

$$0 < x < y < 1$$

$$\text{So, } 0 < x < 1, \quad x < y < 1$$

marginal distribution of X :

$$f_x(x) = \int_x^1 2 dy = 2 \left[y \right]_x^1 = 2(1-x)$$

marginal distribution of Y :

$$f_y(y) = \int_0^y 2 dx = 2 \left[x \right]_0^y = 2[y-0] = 2$$

conditional distribution:

$$f(x|y) = \frac{f(x,y)}{f_y(y)} = \frac{2}{2y}$$

$$f(y|x) = \frac{f(x,y)}{f_x(x)} = \frac{2}{(1-x)}$$

00:3

1 **Theorem:**

If X and Y are random variables with joint distribution function F and marginal distribution functions F_X and F_Y , then X and Y are independent if and only if

$$F(x, y) = F_X(x)F_Y(y)$$

For all $(x, y) \in \mathbb{R}^2$.

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6

stating that
***X* and *Y* are independent**
if and only if

$$F(x, y) \equiv F_X(x)F_Y(y)$$

[REDACTED]:

Suppose (X,Y) are a joint distribution with the variances of X and Y **finite** and **positive**.

Denote the mean and variances of X and Y by μ_1, μ_2 and σ_1^2, σ_2^2 , respectively,

and let ρ be the correlation coefficient between X and Y .

07:0

Upon taking the functions $u(\cdot)$ and $v(\cdot)$ to be the identity functions in in the we note that for independent random variables X_1 and X_2 ,

$$E(X_1 X_2) = E(X_1) E(X_2). \quad (1)$$

00:4

9 **Theorem:**

Suppose X_1 and X_2 are independent and that $E(u(X_1))$ and $E(v(X_2))$ exist.

Then

$$E[u(X_1)v(X_2)] = E[u(X_1)]E[v(X_2)].$$

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6

stating that
if X_1 and X_2 are independent then,

$$E[u(X_1)v(X_2)] = E[u(X_1)] E[v(X_2)]$$

provided that the expectations exist.

Binomial Theorem

$$(q + p)^n = \sum_{x=0}^n \binom{n}{x} p^x q^{n-x}$$

01:0

Theorem:

Suppose the joint mgf, $M(t_1, t_2)$, exists for the random variables X_1 and X_2 .

Then, X_1 and X_2 are independent if and only if

$$M(t_1, t_2) = M(t_1, 0)M(0, t_2);$$

That is, the joint mgf is identically equal to the product of the marginal mgfs.

00:4

9

Theorem stating that X_1 and X_2 are independent
if and only if

**the joint MGF is identically equal to
the product of the marginal MGFs**

i.e.

$$M(t_1, t_2) = M(t_1, 0) M(0, t_2)$$

Example:

For the normal distribution, the **odd** order moments about the mean are all zero and the **even** order moments are given by

$$\mu_{2n} = (2n-1)(2n-3)\dots 5.3.1 \sigma^{2n}$$

00:2

0

██████████:

Let $X_1, X_2, \dots, X_m$ be **independent** random variables such that X_i has binomial $b(n_i, p)$ distribution, for $i = 1, 2, \dots, m$.

Let,
$$Y = \sum_{i=1}^m X_i$$

Then Y has a binomial $b\left(\sum_{i=1}^m n_i, p\right)$ distribution.

00:0

8

**The sum of m
independent
Binomial random variables
with same value of p
is also Binomial**

Q. - Ques.

Derive the CV of exponential distribution?

Answer:

The coefficient of variation is $CV = \frac{\sigma}{\mu}$,

Hence the exponential distribution is given as

$$f(x) = de^{-dx}, \quad 0 < x < \infty$$

the coefficient of variation "CV" is

$$CV = \frac{\sigma}{\mu} = \frac{1/d}{1/d} = d - d$$

Ans

Paper: STA 642.

PDF of the exponential distribution with

mean = $\theta = 0.5, 1, 1.5$.

Answer:

when $\theta = 0.5$

$$f(x) = \left[\frac{1}{0.5} \right] e^{-\left[\frac{x}{0.5} \right]}, \quad 0 < x$$

when: $\theta = 1$

$$f(x) = \left[\frac{1}{1} \right] e^{-\left[\frac{x}{1} \right]}$$

$$f(x) = 1 e^{-x} = e^{-x}$$

when $\theta = 1.5$

$$f(x) = \left[\frac{1}{1.5} \right] e^{-\left[\frac{x}{1.5} \right]}, \quad 0 < x < \infty$$

00:2

4 XXXXXXXXXX:

If X and Y have a bivariate normal distribution with correlation coefficient ρ . Then X and Y are independent **if and only if** $\rho = 0$.

Two minor accidents occur at a busy crossing. Per day, what is the probability at the most two minor accidents will occur during an entire week?

Solution:

This is a poisson process with occurrence rate $\lambda = 2$ also.

one week = 7 days implies $t = 7$

Hence,

So, that,

$$\lambda t = 2 \times 7 = 14$$

$$P(X=x) = \begin{cases} \frac{e^{-\lambda t} (\lambda t)^x}{x!} = \frac{e^{-14} (14)^x}{x!} \\ 0 \text{ elsewhere,} \end{cases}$$

$$x = 0, 1, 2, \dots$$

Now, we require $X \leq 2 \rightarrow (1)$

So,

$$P(X \leq 2) = P(X=0, X=1 \text{ or } X=2)$$

$$= \frac{e^{-14} (14)^0}{0!} + \frac{e^{-14} (14)^1}{1!} + \frac{e^{-14} (14)^2}{2!}$$

$$= e^{-14} + 14e^{-14} + 98e^{-14} = 113e^{-14}$$

$$= \frac{113}{1202592.95} = 0.00009396$$

Q

5 marks.

Derive Mgf of a binomial distribution?

Answer:

The Mgf of binomial distribution:

$$M(t) = \sum_x e^{tx} P(x)$$

$$= \sum_{x=0}^n e^{tx} \binom{n}{x} P^x (1-P)^{n-x}$$

$$= \sum_{x=0}^n \binom{n}{x} (e^t P)^x (1-P)^{n-x}$$

$$= [(1-P) + Pe^t]^n$$

For all real values of t .

general form:

$$M_x(t) = (1-P + Pe^t)^n$$

5 marks.

Derive Mgf of a binomial distribution?

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For All real values of t .

general form:

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Probability density function of Beta distribution is:

$$f(x) = \frac{x^{a-1} (1-x)^{b-1}}{B(a, b)}, \quad 0 \leq x \leq 1$$

$a > 0, b > 0$

Find Probability density of beta distribution when $\alpha = 2, \beta = 5$.

Solution:

By formula: $f(x) = \frac{x^{a-1} (1-x)^{b-1}}{B(a, b)}$

$$f(x) = \frac{x^{2-1} (1-x)^{5-1}}{B(2, 5)},$$

Now solving Beta function

$$B(a, b) = \frac{\Gamma(a) \Gamma(b)}{\Gamma(a+b)}$$

$$B(2, 5) = \frac{\Gamma(2) \Gamma(5)}{\Gamma(2+5)} = \frac{\Gamma(2) \Gamma(5)}{\Gamma(7)}$$

$$= \frac{(2-1)! (5-1)!}{(7-1)!}$$

$$= \frac{(1)! (4)!}{(6)!} = 0.03333$$

standardization formula

$$Z = \frac{X - \mu}{\sigma}$$

Percentile of normal distribution:

$$X = \mu + \sigma Z$$

where μ is mean.

σ is standard deviation
of the variable X .

Z is the standard normal
distribution of desired percent

Let X be $N(2, 25)$ and suppose that we compute the probability

$$P(0 < X < 10) = F(10) - F(0) ?$$

Solution:

Apply Formula $z = \frac{X - \mu}{\sigma} = \frac{X - 2}{5}$

we obtain

$$P(0 < X < 10) = F(10) - F(0)$$

$$= \Phi\left[\frac{10-2}{5}\right] - \Phi\left[\frac{0-2}{5}\right]$$

$$= \Phi\left[\frac{8}{5}\right] - \Phi\left[-\frac{2}{5}\right]$$

$$= \Phi(1.60) - \Phi(-0.40)$$

~~using the standard normal distribution table~~

$$= 0.945 - (1 - 0.655)$$

$$= 0.600 = 60\%$$

An event has the probability $p = \frac{3}{8}$
Find the complete binomial distribution for $n = 5$ trial?

Answer:

$$\text{Here } p = \frac{3}{8}, n = 5$$

$$q = ? \quad q = 1 - p = 1 - \frac{3}{8}$$

$$q = \frac{8-3}{8} = \frac{5}{8}$$

Hence, the desired probabilities
in the binomial expansion of
 $(\frac{5}{8}, \frac{3}{8})^5$.

$$\left[\binom{5}{0} \left(\frac{5}{8}\right)^5 \left(\frac{3}{8}\right)^0 + \binom{5}{1} \left(\frac{5}{8}\right)^4 \left(\frac{3}{8}\right)^1 + \binom{5}{2} \left(\frac{5}{8}\right)^3 \left(\frac{3}{8}\right)^2 + \binom{5}{3} \left(\frac{5}{8}\right)^2 \left(\frac{3}{8}\right)^3 + \binom{5}{4} \left(\frac{5}{8}\right)^1 \left(\frac{3}{8}\right)^4 + \binom{5}{5} \left(\frac{5}{8}\right)^0 \left(\frac{3}{8}\right)^5 \right]$$

x 0 1 2 3 4 5 θ

$$P(x) = (x) 0.09 \quad 0.28 \quad 0.32 \quad 0.20 \quad 0.007$$

The BMI for men aged 60 is normally distributed with mean equal to 29. with standard deviation 6 and z-value is 1.28?

Answer :

To compute the 90th percentile we use the formula

$$X = \mu + \sigma z \rightarrow (1)$$

$$X = ? \quad \mu = 29, \quad \sigma = 6$$

$$z = 1.28$$

$$X = 29 + 6(1.28)$$

$$= 29 + 7.68$$

$$= 36.68 \quad \text{Ans.}$$

Let X denote the number that we observe on the uppermost face if we roll a fair die once?

Answer

Then we have,

X	1	2	3	4	5	6
$P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Suppose we roll a fair die one time and want cdf distribution in mathematics and algebraic form.

Answer:

cumulative distribution:

X	1	2	3	4	5	6
$P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
relative	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	$\frac{6}{6}$

write as

$$f = \begin{cases} \frac{1}{6} & -\infty \leq x < 1 \\ \frac{2}{6} & 1 \leq x < 2 \\ \frac{3}{6} & 2 \leq x < 3 \\ \frac{4}{6} & 3 \leq x < 4 \\ \frac{5}{6} & 4 \leq x < 5 \\ \frac{6}{6} & 5 \leq x < 6 \\ \frac{6}{6} & 6 \leq x < \infty \end{cases}$$

The BMT for men aged 60 is normally distributed with mean equal to 29. with standard deviation 6. and z-value is 1.28?

Answer :

To compute the 90th percentile we use the formula

$$X = \mu + \sigma z \rightarrow (1)$$

$$X = ? \quad \mu = 29, \quad \sigma = 6$$

$$z = 1.28$$

$$X = 29 + 6(1.28)$$

$$= 29 + 7.68$$

$$= 36.68 \quad \text{Ans.}$$

Find probability for a planted rose seed germinates is 0.80. A gardener plants nine seeds?

What is the average number of seeds the gardener could expect to germinate?

What is the standard deviation?

Answer:

The gardener could expect

$$= 9 \times 0.80 = 7.2 \text{ seeds.}$$

Variance of X :

$$np(1-p) = 9 \times 0.80 \times 0.2$$

$$= 1.44$$

The standard deviation of X is the square root of 1.44

Suppose that a game consists of
tossing a fair coin until we
obtain the 3rd head,

$$p = P(H) = 0.5 \text{ and } r = 3.$$

$$P(X=x) = \binom{x-1}{3-1} (0.5)^3 (1-0.5)^{x-3}$$

$$x = 3, 3+1, 3+$$

$$P(X=x) = \binom{x-1}{2} (0.5)^x, \quad x = 3, 4, 5$$

$$x = 3$$

$$P(X=3) = \binom{3-1}{2} (0.5)^3$$

$$= \binom{2}{2} (0.5)^3 = 0.1250$$

$$P(X=4) = \binom{4-1}{2} (0.5)^4$$

$$= \binom{3}{2} (0.5)^4 = 0.1875$$

$$P(X=5) = \binom{5-1}{2} (0.5)^5$$

$$= \binom{4}{2} (0.5)^5 = 0.1875$$

9.6 the mgf of a random variable X is

$$\text{Answer: } M(t) = \left[\frac{2}{3} + \frac{1}{3} e^t \right]^5 ?$$

Then, X has a binomial distribution with $n = 5$, $p = \frac{1}{3}$ that is the PMF of X is

$$P(x) = \begin{cases} \binom{5}{x} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{5-x} \\ 0 \end{cases}, x = 0, 1, \dots, 5$$

$$\mu = np = (5)\left(\frac{1}{3}\right) = \frac{5}{3}$$

and

$$\sigma^2 = np(1-p)$$

$$= 5\left(\frac{1}{3}\right)\left(1 - \frac{1}{3}\right)$$

$$= \frac{5}{3}\left(\frac{3-1}{3}\right) = \frac{5}{3}\left(\frac{2}{3}\right) = \frac{10}{9}$$

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Standard deviation of the discrete Uniform distribution:

$$\sigma = \sqrt{\frac{n^2 - 1}{12}}$$

Let us consider the rolling of die when X is representing the number of dots on uppermost face of the die, the values would be 1, 2, 3, 4, 5, 6? ~~1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12~~

Answer:

$n = 6$, probabilities are $1/6$

mean = ?

$$\text{mean} = E(X) = \mu = \frac{n+1}{2}$$

$$\mu = \frac{6+1}{2} = 3.5$$

$$f(x, y) = \begin{cases} 2 & 0 < x < y, 0 < y < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find the marginal distribution?

Answer:

• Marginal distribution of X :

$$f_x(x) = \int_x^1 2 dy = 2 \left[y \right]_x^1 = 2[1-x]$$

$$= 2(1-x), \quad 0 < x < 1$$

and

marginal distribution of y :

$$f_y(y) = \int_0^y 2 dx = 2 \left[x \right]_0^y = 2(y-0)$$

$$= 2y, \quad 0 < y < 1$$

A game consists of tossing a fair coin until we obtain the first head and $P = P(H) = 0.5$

Answer:

Geometric distribution is given by

$$P_X(x) = (1-p)^{x-1} (p), \quad x=1, 2, 3, \dots$$

$$P_X(x) = (1-0.5)^{x-1} (0.5)$$

$$= (0.5)^{x-1} (0.5)$$

$$= (0.5)^{x-1+1}$$

$$= (0.5)^x \quad \because 0.5 = \frac{1}{2}$$

$$P_X(x) = \left(\frac{1}{2}\right)^x$$

$$x=1$$

$$P_X(1) = \left(\frac{1}{2}\right)^1 = \frac{1}{2} = 0.5$$

$$x=2$$

$$P_X(2) = \left(\frac{1}{2}\right)^2 = \frac{1}{4} = 0.25$$

$$x=3$$

$$P_X(3) = \left(\frac{1}{2}\right)^3 = \frac{1}{8} = 0.125$$

$$P_X(4) = \left(\frac{1}{2}\right)^4 = \frac{1}{16} = 0.0625 \quad x=4$$

$$\frac{1}{16} = 0.0625$$

P

An oil company conducts have 20% chance of striking oil. The company wants of success twice, i.e. Two oil wells. What is the probab. that the second success comes on fifth well drilled?

Solution:

$$P = 20\% = 0.2, \quad r = 5, \quad \delta = 2$$

$$P(X=r) = \left[\begin{matrix} r-1 \\ \delta-1 \end{matrix} \right] P^{\delta} (1-P)^{r-\delta}$$

$$r = \delta, \delta+1, \delta+2, \dots$$

Putting values.

$$P(X=5) = \left[\begin{matrix} 5-1 \\ 2-1 \end{matrix} \right] (0.2)^2 (1-0.2)^{5-2}$$

$$= \left[\begin{matrix} 4 \\ 1 \end{matrix} \right] (0.2)^2 (0.8)^3$$

$$= \left[\begin{matrix} 4 \\ 1 \end{matrix} \right] (0.2)^2 (0.8)^3$$

$$= 4(0.2)^2 (0.8)^3$$

$$= 0.082$$

$$= 8.2\% \quad \underline{P}$$

4 Definition:

A random variable X is said to have a normal distribution if its pdf is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty.$$

- μ is the mean or expectation of the distribution (and also its median and mode),
 - $\sigma > 0$ is the standard deviation, and
 - σ^2 is the variance.
-

4 Cumulative Distribution Function

5 The formula for the **cumulative distribution** function of the normal distribution is

where

$$F(x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x-\mu}{\sigma\sqrt{2}} \right) \right] \quad \text{for } -\infty < x < \infty.$$

When do we apply this formula?

14:2

Poisson distribution arrives in two situations:

1. It appears as the limiting form of the binomial distribution under certain conditions.
2. When we apply the  that real-life situation where we are dealing with Poisson process

Define incomplete gamma function:

$$I_f F(x) = \begin{cases} \frac{1}{\Gamma(\alpha)} \gamma\left(\alpha, \frac{x}{\beta}\right) & \alpha > 0, \beta > 0 \\ 0 & \end{cases}$$

where $\gamma\left(\alpha, \frac{x}{\beta}\right)$ is the incomplete gamma function

The upper incomplete gamma function is defined as:

$$\Gamma(\alpha, y) = \int_y^{\infty} t^{\alpha-1} e^{-t} dt \quad \text{S.O. +}$$

The lower incomplete gamma function is defined as:

$$\gamma(\alpha, y) = \int_0^y t^{\alpha-1} e^{-t} dt$$

it is interesting to note that

$$\gamma(\alpha, y) + \Gamma(\alpha, y) = \Gamma(\alpha)$$



7

Derivation ofthe **mean****Distribution****00:2****0**A very interesting property:

- The hazard rate of this distribution is constant.
- The **mean and standard deviation** of the exponential distribution are equal.

00:5**6**1. Derivation of the **mean**:

The pdf of the Exponential distribution is given by

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \lambda > 0.$$

Therefore,

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \lambda e^{-\lambda x} dx$$

02:1**00:2****0**A very interesting property:

- The hazard rate of this distribution is constant.
- The **mean and standard deviation** of the exponential distribution are equal.

9Example 1:

Suppose that X has a **Poisson distribution** with $\lambda = 4$.

Then the pmf of X is

$$p(x) = \begin{cases} \frac{0.5^x e^{-0.5}}{x!} & x = 0, 1, 2, \dots \\ 0 & \text{elsewhere.} \end{cases}$$

05:0

0The mean of this distribution is 4.

&

The variance of this distribution is 4.

Therefore,

The **coefficient of variation** of this distribution is

$$COV = \frac{\sigma}{\mu} = \frac{\sigma}{\mu} \times 100\% = \frac{\sqrt{4}}{4} \times 100\% = \frac{1}{2} \times 100\% = 50\%$$

05:1
5

cumulative distribution Function

- The formula for the **cumulative distribution** function of the beta distribution is also called the **incomplete β function ratio** (commonly denoted by I_x) and is defined as

$$F(x) = I_x(\alpha, \beta) = \frac{\int_0^x t^{\alpha-1} (1-t)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq x \leq 1; \alpha, \beta > 0$$

where B is the beta function defined above.

Formula of Variance =

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \mu^2 + \sigma^2 - \mu^2$$

$$= \sigma^2$$

Formula of Variance of the Hypergeometric distribution

$$\text{Variance} = n \frac{K}{N} \cdot \frac{(N-K)}{N} \cdot \frac{N-1}{N-1}$$

Two properties of normal curve:

1. The total area (probability) under the normal curve is unity.

2. The graph of the normal curve is bell-shaped.

3. The normal curve has the zero kurtosis.

$$f(x_1, x_2) = \begin{cases} x_1 + x_2 & 0 < x_1 < 1, 0 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

show that X_1 and X_2 are dependent?

Marginal probability of x_1 :

$$f_1(x_1) = \int_0^1 (x_1 + x_2) dx_2$$

$$= \left[x_1 x_2 + \frac{x_2^2}{2} \right]_0^1$$

$$= x_1(1) - x_1(0) + \frac{(1)^2}{2} - \frac{0}{2}$$

$$f(x_1) = x_1 + \frac{1}{2}, \quad 0 < x_1 < 1$$

Marginal probability of x_2 :

$$f_2(x_2) = \int_0^1 (x_1 + x_2) dx_1$$

$$= \left[\frac{x_1^2}{2} + x_1 x_2 \right]_0^1$$

$$= \frac{(1)^2}{2} - \frac{0}{2} + (1)x_2 - (0)(x_2)$$

$$f(x_2) = \frac{1}{2} + x_2$$

$$\text{Now } f_1(x_1)f_2(x_2) = \left(\frac{1}{2} + x_1\right)\left(x_2 + \frac{1}{2}\right)$$

$$= \frac{1}{2}x_2 + \frac{1}{4} + x_1x_2 + \frac{1}{2}x_1$$

$$= x_1x_2 + \frac{1}{2}x_1 + \frac{1}{2}x_2 + \frac{1}{4}$$

$$f(x_1, x_2) = (x_1 + x_2), \quad f(x_1, x_2) \neq f_1(x_1)f_2(x_2) \text{ so,}$$

X_1 and X_2 are dependent.

Binomial Experiment

00:5

0 An experiment is called a **binomial experiment** if it possesses the following **four** properties:

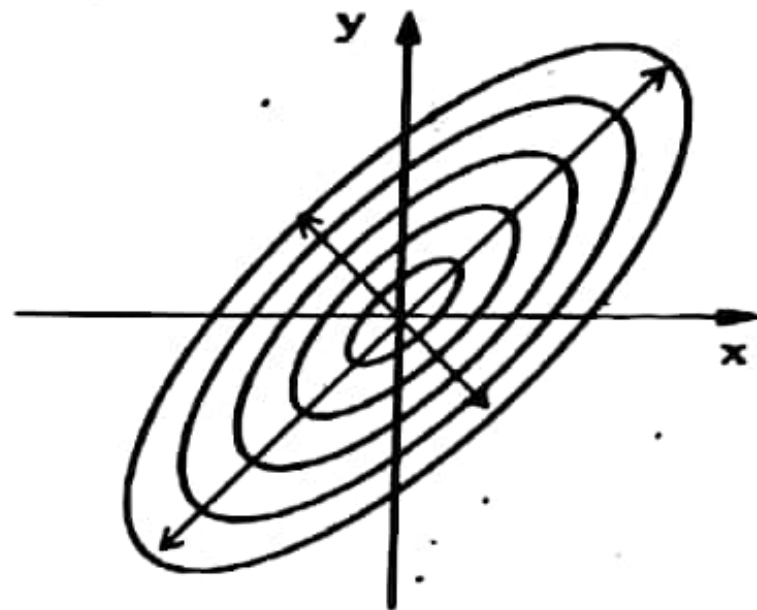
1. The outcome of each trial may be classified into one of two categories, conventionally called **success (S)** and **Failure (F)**.
-

13:47

General Concept of **Ellipse**:

We may assume that **one** of the many diameters of the circle is being 'elongated'.

As soon as we
'elongate' one of the
diameters, we get a
major axis as well as
a minor axis.

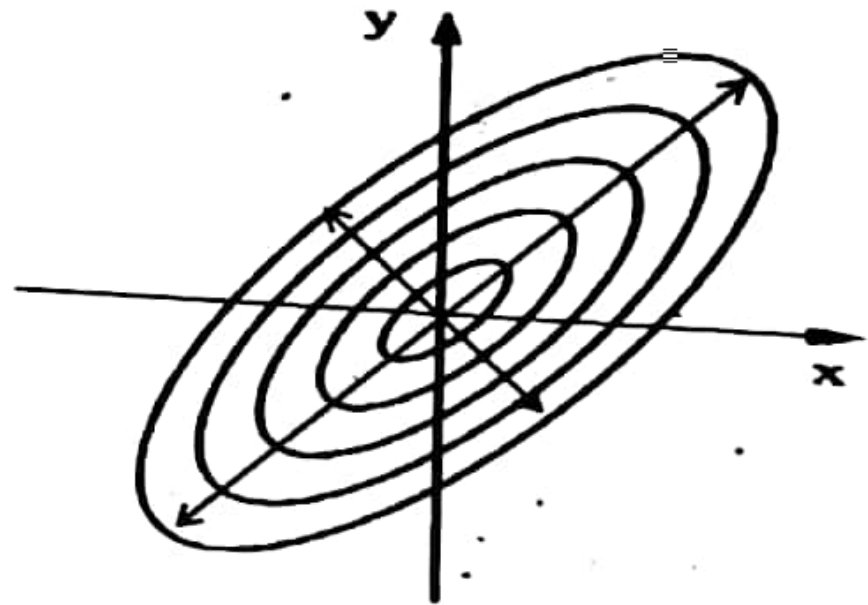


13:47

General Concept of [REDACTED]

We may assume that **one** of the many diameters of the circle is being 'elongated'.

As soon as we
'elongate' one of the
diameters, we get a
major axis as well as
a minor axis.



formula of variance discrete uniform distribution

$$\sigma = \sqrt{\frac{n^2 - 1}{12}}$$

03:1

8

Cumulative Distribution Function

write formula of cdf continuous uniform distribution

The Cumulative Distribution function of the **continuous uniform** distribution is:

$$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b, \\ 1 & \text{for } x > b \end{cases}$$

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

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write pdf of normal distribution of random variable x

A random variable X is said to have a normal distribution if its pdf is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

where μ is the mean of the distribution and $\sigma > 0$ is the standard deviation.

We say that X has the $N(\mu, \sigma^2)$ distribution.

variance coefficient of variation of poisson distribution

Explanation:

The **coefficient** of variation is defined as $CV = \frac{\sigma}{\mu}$.

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The name of 5 men and 5 women are written on slips of paper and placed in a hat. 4 names are drawn what is the probability that 2 are men and 2 are women?

Solution:

Let X denote the number of men then,

$$N = 5 + 5 = 10 \text{ names}$$

$$K = 5, \text{ and } n = 4$$

The ~~hyper~~ hypergeometric distribution is:

$$h(x; 10, 4, 5) = \frac{\binom{5}{x} \binom{5}{4-x}}{\binom{10}{4}}$$

Probability is 2, $x = 2$

$$h(2; 10, 4, 5) = \frac{\binom{5}{2} \binom{5}{4-2}}{\binom{10}{4}}$$

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Rotate



Markup



Word



Signature



Note

5 marks.

Write simple general first derivative of normal distribution?

Solution:

We know the normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}$$

By taking first derivative

$$f'(x) = \frac{d}{dx} \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2} \right]$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \frac{d}{dx} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}$$

$$= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2} \frac{d}{dx} \left[-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2 \right]$$

$$= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2} \cdot \left[\left(-\frac{1}{2}\right) \cdot \left[\frac{1}{\sigma}\right] \cdot \left[\frac{x-\mu}{\sigma}\right] \right]$$

$$= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2} \cdot \left[-\left[\frac{x-\mu}{\sigma^2}\right] \right]$$

$$f'(x) = -\left[\frac{x-\mu}{\sigma^3}\right] \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}$$

$$= -\frac{1}{\sigma^3\sqrt{2\pi}} (x-\mu) \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

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